**Additional file 2**

**Table 2. Study characteristics**

| Author (year) | Level | Diagnosis | Phase stroke | Patients/ controls (n) | Age group | Reference test | Measured joints | Device name | Measurement | Measurement outcome | Test retest |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Piovesan et al. (2011) | 0 | Stroke | Chronic | 9/0 | Adult | AS | Shoulder + elbow | Braccio di Ferro | Spasticity | Resistance torque | No |
| Stienen et al. (2011) | 0 | Stroke | NA | 1/0 | NA | NA | Elbow | ACT-4D HapticMaster | Spasticity | Torque + EMG | No |
| Piovesan et al. (2011) | 0 | Stroke | Chronic | 9/0 | Adult | AS | Shoulder + elbow | Braccio di Ferro | Spasticity + Viscoelastic joint properties | Torque Angle Reaction | No |
| Zadravec et al. (2014) | 0 | Stroke | Chronic | 1/0 | Adult | MAS | Elbow | Haptic Master robot system | Spasticity | Torque Angle Reaction | No |
| Godfrey et al. (2010) | 0 | Stroke | Chronic | 4/0 | Adult | AS | Fingers | HEXORR | Viscoelastic joint properties | Torque Angle Reaction + PROM | No |
| Ren et al. (2013) | 0 | Stroke | Chronic | 3/2 | Adult | NA | Shoulder + elbow + wrist | IntelliArm | Viscoelastic joint properties | PROM | No |
| Schmartz et al. (2011) | 0 | CP | NA | 9/0 | Child | NA | Hip + knee | L-STIFF | Spasticity + Viscoelastic joint properties | Torque Angle Reaction | Yes |
| Palazzolo et al. (2007 | 0 | Stroke | NA | 1/0 | Elderly | MAS | Shoulder + elbow | MIT-MANUS | Spasticity | Impedance | No |
| Posteraro et al. (2018) | 0 | Stroke | Chronic | 5/0 | Adult | MAS | Elbow | NEUROExos Elbow Module system | Spasticity | Torque Angle Reaction | No |

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|---|---|---|---|---|---|---|---|---|---|---|---|
| Zhoa et al. (2011) | 0 | CP | NA | 7/0 | Child | NA | Ankle | No Name | Viscoelastic joint properties | Torque Angle Reaction | No |
| **Author (year)** | **Level** | **Diagnosis** | **Phase stroke** | **Patients/ controls (n)** | **Age group** | **Reference test** | **Measured joints** | **Device name** | **Measurement** | **Measurement outcome** | **Test retest** |
| Zhang et al. (2002) | 0 | Stroke | Chronic | 4/5 | Adult | NA | Ankle | No Name | Spasticity + Viscoelastic joint properties | Torque Angle Reaction + PROM + EMG | No |
| Lerner et al. (2019) | 0 | CP | NA | 6/0 | Child | MAS | Knee | No Name | Spasticity | Stretch-reflex velocity gain | No |
| Park et al. (2008) | 0 | Stroke | Chronic | 4/0 | Adult | NA | Elbow | No Name | Spasticity + Viscoelastic joint properties | Torque Angle Reaction | No |
| Kung et al. (2016) | 0 | Stroke | Chronic | 9/0 | Adult | MAS | Shoulder + elbow | No Name | Viscoelastic joint properties | Force Angle Reaction + Passive ROM | No |
| Zhou et al. (2016) | 0 | Stroke | Chronic | 7/7 | Adult | NA | Ankle | PKU-RARS | Viscoelastic joint properties | Torque Angle Reaction + PROM | No |
| Ranzani et al. (2019) | 0 | Stroke | Chronic | 5/5 | Adult | MAS | Fingers | ReHapticKnob | Viscoelastic joint properties | Torque Angle Reaction | No |
| Kim et al. (2013) | 0 | Stroke | Chronic | 1/0 | Adult | NA | Knee | Walkbot system | Viscoelastic joint properties | Torque Angle Reaction | No |
| Baniasad et al. (2011) | 0 | Stroke | Chronic | 1/0 | Adult | MAS | Wrist | Wrist-RoboHab | Spasticity | Torque Angle Reaction | No |
| McPherson et al. (2019) | I | Stroke | Chronic | 13/0 | Adult | MAS | Elbow | Biodex Rehabilitation testing system | Spasticity + Viscoelastic joint properties | Torque Angle Reaction | No |

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|---|---|---|---|---|---|---|---|---|---|---|---|
| Xu et al. (2020) | I | CP | NA | 17/17 | Child | NA | Ankle | IntelliStretch | Spasticity + Viscoelastic joint properties | Torque Angle Reaction + EMG | No |
| Wang et al. (2017) | I | Stroke | Chronic | 17/17 | Adult | NA | Wrist | Neuroflexor | Spasticity + Viscoelastic joint properties | Resistance torque | No |

| **Author (year)** | **Level** | **Diagnosis** | **Phase stroke** | **Patients/ controls (n)** | **Age group** | **Reference test** | **Measured joints** | **Device name** | **Measurement** | **Measurement outcome** | **Test retest** |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Gaverth et al. (2013) | I | Stroke | Chronic | 34/10 | Adult | PROM + MAS | Wrist + fingers | Neuroflexor | Spasticity + Viscoelastic joint properties | Resistance torque | Yes |
| Kamper et al. (2000) | I | Stroke | Chronic | 13/2 | Adult | NA | Fingers | No Name | Spasticity + Viscoelastic joint properties | Torque Angle Reaction + PROM + EMG | No |
| Kamper et al. (2003) | I | Stroke | Chronic | 12/1 | Adult | NA | Fingers | No Name | Spasticity | Torque Angle Reaction; + EMG | No |
| Kamper et al. (2006) | I | Stroke | Chronic | 30/0 | Adult | NA | Fingers | No Name | Spasticity + Viscoelastic joint properties | Torque Angle Reaction + EMG | No |
| Sin et al. (2019) | I | Stroke | Chronic | 17/0 | Adult | MAS + MTS | Elbow | No Name | Spasticity | Torque Angle Reaction + EMG | Yes |
| de Gooijer-Van de Groep et al. (2018) | I | Stroke | Subacute | 36/0 | Adult | NA | Wrist | Wristalyzer | Spasticity + Viscoelastic joint properties | Torque Angle Reaction | No |
| Van der Krogt et al. (2015) | I | Stroke | Chronic | 32/14 | Adult | NA | Wrist | Wristalyzer | Spasticity + Viscoelastic joint properties | Torque Angle Reaction + ROM + reflex loop time | Yes |

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|---|---|---|---|---|---|---|---|---|---|---|---|
| Germanotta et al. (2020) | II | Stroke | Subacute | 120/0 | Adult | MAS | Fingers | Amadeo | Spasticity | Torque Angle Reaction + PROM | Yes |
| Roy et al. (2011) | II | Stroke | Chronic | 10/20 | Adult | MAS + ROM | Ankle | Anklebot | Viscoelastic joint properties | Torque Angle Reaction | No |
| Starsky et al. (2005) | II | Stroke | Chronic | 16/0 | Adult | MAS | Elbow | Biodex Multi-Joint System 3B | Spasticity + Viscoelastic joint properties | Torque Angle Relation | Yes |

| Author (year) | Level | Diagnosis | Phase stroke | Patients/ controls (n) | Age group | Reference test | Measured joints | Device name | Measurement | Measurement outcome | Test retest |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Centen et al. (2017) | II | Stroke | Chronic | 46/96 | Adult | MAS + MTS | Elbow | KINARM robotic exoskeleton | Spasticity + Viscoelastic joint properties | Torque Angle Reaction + PROM + release after catch | No |
| Leng et al. (2019) | II | Stroke | Subacute | 15/0 | Adult | NA | Wrist | Neuroflexor | Spasticity + Viscoelastic joint properties | Resistance torque | No |
| Wang et al. (2018) | II | Stroke | Chronic | 21/0 | Adult | NA | Wrist | Neuroflexor | Spasticity + Viscoelastic joint properties | Resistance torque | No |
| Gaverth et al. (2014) | II | Stroke | Chronic | 22/0 | Adult | PROM + MAS | Wrist + fingers | Neuroflexor | Spasticity + Viscoelastic joint properties | Resistance torque | No |
| Wu et al. (2010) | II | CP | NA | 12/0 | Child | MAS | Ankle | No Name | Viscoelastic joint properties | Torque Angle Reaction + PROM | No |
| Chen et al. (2014) | II | CP | NA | 12/0 | Child | MAS + MTS | Ankle | No Name | Viscoelastic joint properties | PROM | No |
| Willerslev-Olsen et al. (2013) | II | CP | NA | 35/28 | Child | MAS + MTS | Ankle | No Name | Spasticity + Viscoelastic joint properties | Torque response to stretch + EMG | No |
| Gao et al. (2011) | II | Stroke | Chronic | 10/10 | Adult | NA | Ankle | No Name | Viscoelastic joint properties | Torque Angle Reaction | No |
| Waldman et al. (2013) | II | Stroke | Chronic | 12/12 | Adult | MAS | Ankle | No Name | Viscoelastic joint properties | PROM | No |
| Chen et al. (2016) | II | CP | NA | 41/0 | Child | MAS | Ankle | No Name | Viscoelastic joint properties | PROM | No |

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|---|---|---|---|---|---|---|---|---|---|---|---|
| Chung et al. (2004) | II | Stroke | Chronic | 24/32 | Adult | MAS | Ankle | No Name | Viscoelastic joint properties | Torque Angle Reaction + PROM | No |

| **Author (year)** | **Level** | **Diagnosis** | **Phase stroke** | **Patients/ controls (n)** | **Age group** | **Reference test** | **Measured joints** | **Device name** | **Measurement** | **Measurement outcome** | **Test retest** |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Selles et al (2005) | II | Stroke | Chronic | 10/0 | Adult | NA | Ankle | No Name | Spasticity + Viscoelastic joint properties | Torque Angle Reaction + PROM | No |
| Lorentzen et al. (2010) | II | Stroke | Chronic | 10/31 | Adult | AS | Ankle | No Name | Spasticity + Viscoelastic joint properties | Torque Angle Reaction + ROM + EMG | No |
| de Vlugt et al. (2010) | II | Stroke | Chronic | 19/7 | Adult | AS | Ankle | No Name | Spasticity + Viscoelastic joint properties | Torque Angle Reaction + ROM | Yes |
| Nam et al. (2017) | II | Stroke | Chronic | 20/1 | Adult | MAS | Elbow + wrist | No Name | Spasticity | Torque Angle Reaction | Yes |
| Fischer et al. (2007) | II | Stroke | Chronic | 15/0 | Adult | NA | Fingers | No Name | Spasticity + Viscoelastic joint properties | Torque Angle Reaction + PROM | No |
| Chen et al. (2014) | II | CP | NA | 23/0 | Child | MAS | Ankle | No Name | Spasticity + Viscoelastic joint properties | PROM | No |
| Xiadoyan et al. (2017) | II | Stroke | Chronic | 14/0 | Adult | MAS | Wrist | No Name | Spasticity + Viscoelastic joint properties | Torque Angle Reaction + PROM | No |
| Wu et al. (2011) | II | CP | NA | 12/0 | Child | MAS + MTS | Ankle | No Name | Viscoelastic joint properties | Torque Angle Reaction | No |
| Wu et al. (2010) | II | CP | NA | 12/0 | Child | MAS | Ankle | No Name | Viscoelastic joint properties | Torque Angle Reaction + PROM | No |

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|---|---|---|---|---|---|---|---|---|---|---|---|
| Lum et al. (1999) | II | Stroke | Chronic | 13/6 | Adult | NA | Elbow | PUMA 260 robot | Spasticity | Torque Angle Reaction | Yes |
| Dehem et al. (2017) | II | Stroke | Chronic | 12/0 | Adult | MAS | Elbow | REAplan robot | Spasticity | Torque Angle Reaction | No |
| De Gooijer-Van de Groep et al. (2016) | II | Stroke | Chronic | 30/14 | Adult | MAS | Wrist | Wristalyzer | Spasticity + Viscoelastic joint properties | Torque Angle Reaction + ROM | No |
| Andringa, A. et al (2020) | II | Stroke | Acute | 17/17 | Adult | MAS | Wrist + fingers | Neuroflexor | Spasticity + Viscoelastic joint properties | Resistance torque | Yes |
| Lindberg, P.G. (2011) | II | Stroke | Chronic | 31/13 | Adult | MAS | Wrist | No Name | Spasticity + Viscoelastic joint properties | Force Angle Reaction + EMG | No |
| Pennati, G.V. et al (2016) | II | Stroke | Acute | 39/107 | Adult | MAS + PROM | Wrist + fingers | Neuroflexor | Spasticity + Viscoelastic joint properties | Resistance torque | Yes |
| Andringa, A. et al (2019) | II | Stroke | Chronic | 45/30 | Adult | MAS + PROM + MTS | Wrist + fingers | Neuroflexor | Spasticity + Viscoelastic joint properties | Resistance torque | Yes |
| Plantin, J. et al (2019) | II | Stroke | Subacute | 55/0 | Adult | PROM + MAS | Wrist + fingers | Neuroflexor | Spasticity + Viscoelastic joint properties | Resistance torque | Yes |

Abbreviations: CP: cerebral palsy; NA: not available; AS: Ashworth Scale; MAS: Modified Ashworth Scale; MTS: Modified Tardieu Scale; (P)ROM: (Passive) Range Of Motion; EMG: Electromyography. Note: Phase stroke: acute 0–7 days; subacute 7 days–3 months; chronic >6 months Age group: child <12 years; young adult 12–18 years; adult 18–65 years; elderly >65 years.